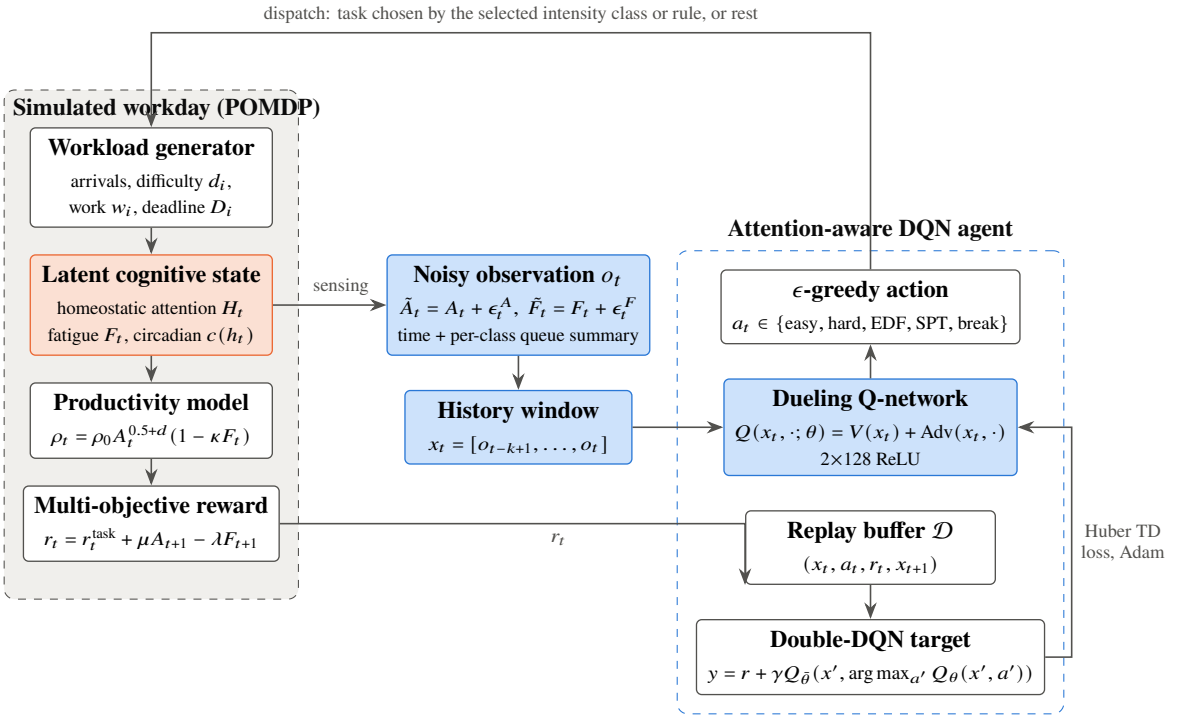

dispatch: task chosen by the selected intensity class or rule, or rest
Simulated workday (POMDP)
Workload generator
arrivals, difficulty d_i,
work w_i, deadline D_i
Latent cognitive state
homeostatic attention H_t
fatigue F_t, circadian $c(h_t)$
Productivity model
$\rho_t = \rho_0 A_t^{0.5+d}(1-\kappa F_t)$
Multi-objective reward
$r_t = r_t^{\text{task}} + \mu A_{t+1} - \lambda F_{t+1}$
sensing
Noisy observation o_t
$\tilde{A}_t = A_t + \epsilon_t^A,\ \tilde{F}_t = F_t + \epsilon_t^F$
time + per-class queue summary
History window
$x_t = [o_{t-k+1}, \ldots, o_t]$
Attention-aware DQN agent
ϵ-greedy action
$a_t \in$ {easy, hard, EDF, SPT, break}
Dueling Q-network
$Q(x_t, \cdot; \theta) = V(x_t) + \text{Adv}(x_t, \cdot)$
2×128 ReLU
r_t
Replay buffer $\mathcal{D}$
(x_t, a_t, r_t, x_{t+1})
Double-DQN target
$y = r + \gamma Q_{\bar{\theta}}(x', \arg\max_{a'} Q_\theta(x', a'))$
Huber TD
loss, Adam